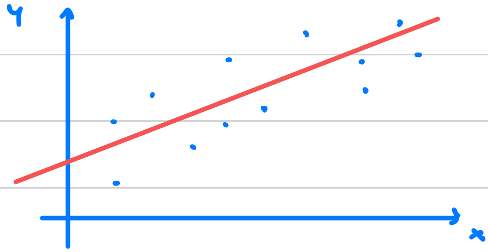
SIMPLE LINEAR REGRESSION ATOMIC SUMMARY



THEORETICAL MODEL

$$y = \beta_0 + \beta_1 x + \varepsilon$$

ESTIMATED MODEL EQUATION (LEAST SQUARES)

$$\hat{y}_i = b_0 + b_1 x_1 \qquad b_1 = \frac{s_{xy}}{s_x^2} = r_{xy}\frac{s_y}{s_x} \qquad b_0 = \bar{y} - b_1\bar{x}$$

ANALYSIS OF VARIANCE : SST = SSR + SSE

$$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2 = s_y^2(n-1) \qquad SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 \qquad SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

COEFFICIENT OF DETERMINATION R^2

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \qquad 0 \le R^2 \le 1$$

represents the proportion of variability of y explained by the model.

ASSUMPTIONS

1. LINEARITY : $E[\varepsilon_i] = 0 \ \forall i$
2. HOMOSCEDASTICITY : $Var(\varepsilon_i) = \sigma_\varepsilon^2 \ \forall i$
3. UNCORRELATED ERRORS : $Cor(\varepsilon_i, \varepsilon_j) = 0 \ \forall i,j$
4. NORMALITY : $\varepsilon_i \sim N(0, \sigma_\varepsilon^2) \ \forall i$

WEAK ASSUMPTIONS (1–3)

STRONG ASSUMPTIONS (1–4)

ESTIMATE s_ε^2 OF σ_ε^2

$$s_\varepsilon^2 = \frac{SSE}{n-2}$$

s_ε is called model/residual standard error

UNDER WEAK ASSUMPTIONS

$$E[\hat{\beta}_0] = \beta_0 \qquad Var(\hat{\beta}_0) = \sigma_\varepsilon^2\left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}\right) \rightarrow \text{optional}$$

$$E[\hat{\beta}_1] = \beta_1 \qquad Var(\hat{\beta}_1) = \frac{\sigma_\varepsilon^2}{(n-1)s_x^2}$$

UNDER STRONG ASSUMPTIONS

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma_\varepsilon^2}{(n-1)s_x^2}\right)$$

optional

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma_\varepsilon^2\left(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}\right)\right)$$

HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

$$s_\varepsilon^2 = \frac{SSE}{n-2} \qquad se(\hat{\beta}_1) = \sqrt{\frac{s_\varepsilon^2}{(n-1)s_x^2}}$$

estimate of standard error of $\hat{\beta}_1$

$$\boxed{\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{s_\varepsilon^2}{(n-1)s_x^2}}} \sim t_{n-2}}$$

CONFIDENCE INTERVAL FOR β_1

$$CI_{0.95}(\beta_1) = b_1 \pm t_{0.025,n-2}\, se(\hat{\beta}_1)$$

CONFIDENCE INTERVAL FOR β_1

TEST FOR THE SIGNIFICANCE OF β_1

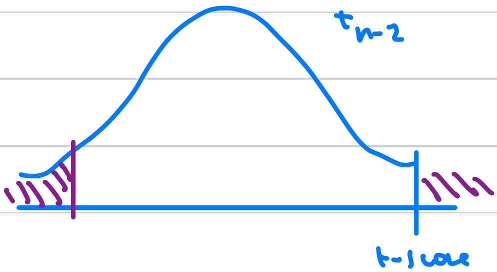
To test the significance of β_1 mean to test whether $H_0: \beta_1 = 0$:

$$\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \longrightarrow t\text{-score} = \frac{b_1}{se(\hat{\beta}_1)}$$

the t-score is a realisation of a t_{n-2} under H_0

$$R_{0.05} = \{t\text{-score} < qt(\overset{\alpha/2}{0.025}, n-2) \text{ or } t\text{-score} > qt(\overset{1-\alpha/2}{0.975}, n-2)\}$$

$$p\text{-value} = [1 - pt(|t\text{-score}|, n-2)] \cdot 2$$



CONFIDENCE INTERVALS FOR FORECASTS GIVEN $x = x_g$

Given $x = x_g$ our point estimate for the forecast is $\hat{y}_g = b_0 + b_1 x_g$

UNDER STRONG ASSUMPTIONS:

- CONFIDENCE INTERVAL FOR THE EXPECTED VALUE OF THE RESPONSE

$$\underline{\mu_g} = \beta_0 + \beta_1 x_g$$

$$\hat{y}_g \pm t_{\alpha/2, n-2} \, s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1) s_x^2}}$$

- PREDICTION INTERVAL FOR THE ACTUAL RESPONSE

$$\underline{y_g} = \beta_0 + \beta_1 x_g + \varepsilon_g$$

$$\hat{y}_g \pm t_{\alpha/2, n-2} \, s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{(n-1) s_x^2}}$$